Süzgeç

Literature, Data and Technology Review

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Süzgeç at a glance

Problem	Users check their phones constantly for fear of missing something important; every notification breaks attention; gambling, shopping-campaign and social media notifications trigger the related addictions.
Target users	Android users aged 18+ with non-flagship phones; university students first.
Core idea	AI sorts notifications into Critical, Important, Normal, Unimportant and Deleted. Critical and Important arrive immediately; the rest are delivered in batches 3 times a day; addiction-driving content (gambling by default, shopping campaigns and social media by user choice) is removed from the notification panel and moved to the Deleted box; users can restore anything. The idea came from the batching experiment of Fitz et al. (2019).
Data	Real notification logs (Komninos 2018/2019), Turkish SMS and e-mail datasets, app usage logs (LSApp), synthetic Turkish notifications and our team's own real notifications.
Approach	On-device hybrid model: rule layer + small Turkish BERT (fallback: TF-IDF + logistic regression) + ranking that learns from user corrections.

Summary: The literature review (8 themes, 52 studies and sources) shows that the problem is checking frequency rather than screen time, that turning off all notifications increases FoMO, and that batched delivery 3 times a day reduces phone unlocks (d = −0.60); importance-aware batching and an addiction filter at the notification level have not been tested. In the data review, across 195,093 notifications from 44 users, half of the notifications were persistent, 26% of the rest disappeared within 5 seconds, and users received a median of 86 notifications per day. In Turkish SMS, campaign language can be separated even with simple rules; in e-mail, the rules break down. The TF-IDF + logistic regression model we tested reached a macro F1 of 0.998 on SMS but did not generalise to another text type (0.62). Based on the technology review, Süzgeç will be built as an on-device hybrid system (rule layer + small Turkish BERT + ranking that learns from corrections).

Part 1 – Literature Review

1.1 Introduction

The problem Süzgeç targets is the frequent, short checking loop rather than long screen time. This section defines the problem, the target users, why the topic matters, and the three questions the literature needs to answer.

Problem: Users check their phones constantly for fear of missing something important (FoMO); every notification breaks attention; gambling, shopping-campaign and social media notifications trigger the related addictions. In this document we use the term "problematic smartphone use" instead of "smartphone addiction", because evidence that the behaviour meets addiction criteria is still insufficient (Panova & Carbonell, 2018).

Target users: Android users aged 18+ with non-flagship phones; university students first. In Turkey, 24.7% of 612 university students showed a possible risk of smartphone addiction (in the study's own terms), and the risk was higher in the 18–24 age group (Akpınar Aslan et al., 2021). In another study with 488 adults, the addiction score of the 51+ group was significantly lower than that of the 20–34 and 35–50 groups (Gökçearslan et al., 2023). 42% of Yeşilay's gambling clients are aged 18–25 (Yeşilay, 2026). Because similar AI notification features are currently offered only on new flagship devices (e.g. Pixel 9 and 10), we focus on mid-range Android devices (details: 3.5).

Why it matters, the "did something come in?" loop: The findings in Table 1 show that checking is frequent, short and mostly unnoticed, and that interruption has a cost independent of screen time. In short, the problem is not duration but interruption and checking frequency.

Table 1. Key findings on the checking loop

Finding	Source	Meaning for Süzgeç
23 people used their phone 85 times a day on average; more than half of the uses were shorter than 30 seconds; people estimated their use at about half of the real value	Andrews et al. (2015)	Checking frequency goes unnoticed; Süzgeç can make it visible in its report
89% of 1,130 phone interactions of 37 people were started by the user, without a notification	Heitmayer and Lahlou (2021)	People check "in case something came in"; the "you will hear it if it is Critical" guarantee targets this empty checking
205 young adults received 151–154 notifications a day on average in the baseline week and checked their phone 85–86 times	Dekker et al. (2025)	The most recent baseline closest to our target users
In 180 students, a notification caused a slowdown lasting about 7 seconds; the slowdown was related not to screen time but to notification volume and checking frequency	Fournier et al. (2026)	The target metric is not screen time but interruptions and number of checks
Participants estimated that they received about 30 notifications a day and named 15 as ideal; in reality they received about 65 (Figure 2)	Fitz et al. (2019)	Users do not notice their notification load; the delivery summary makes it visible

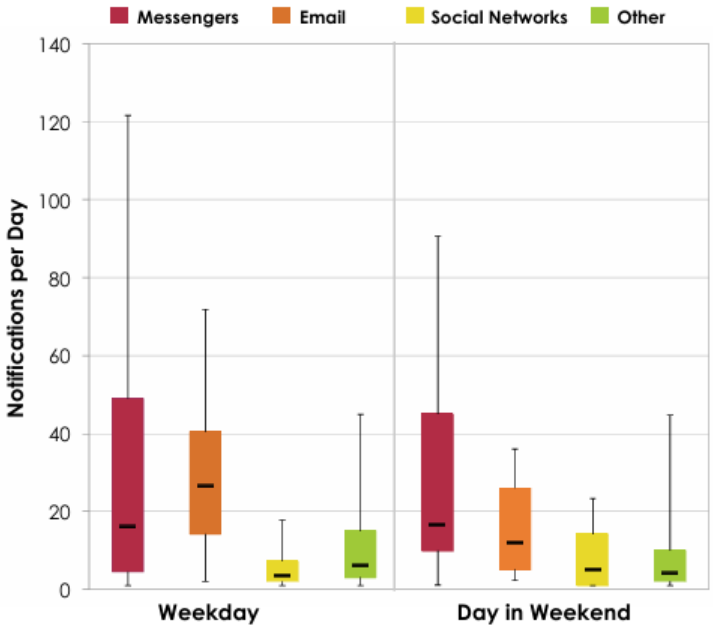


Figure 1. Median number of daily notifications per participant, weekdays and weekends. Source: Pielot et al. (2014), Figure 2.

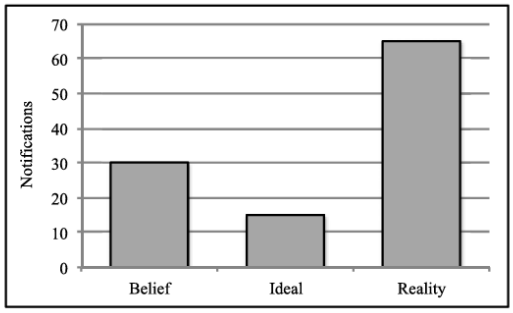


Fig. 5. How many notifications participants think they receive, want to receive, and actually receive.

Figure 2. The number of daily notifications participants believed they received, wanted to receive and actually received. Source: Fitz et al. (2019), Fig. 5.

Why literature first: Each of Süzgeç's core decisions rests on a finding: delivery 3 times a day (Fitz et al., 2019), filtering instead of "turning everything off" (Dekker et al., 2025; Pielot & Rello, 2017), and ranking that learns from user corrections (Lin, Y.-H. et al., 2024). A system designed without the literature could have gone for "turn off all notifications" and increased FoMO.

Research questions: The literature review seeks answers to three questions:

- **RQ1:** How do notifications affect attention and phone-checking behaviour; is the problem screen time, or interruption and checking frequency? (Themes 1–3)
- **RQ2:** Which notification intervention reduces checking without increasing FoMO? (Themes 4, 5 and 8)
- **RQ3:** Can notification importance and addiction-driving content be identified automatically and per person? (Themes 6 and 7)

1.2 Structure

We group the literature into 8 themes according to the research questions and Süzgeç's design decisions (Table 2).

Search method: This is a narrative review, not a systematic review. Searches were run on Google Scholar and publisher databases (ACM Digital Library, ScienceDirect, PubMed Central), on arXiv for preprints, and on DergiPark and TR Dizin for Turkish studies; AI-assisted research tools were used for searching and pre-screening, and every number used was checked against the full text or the publisher's abstract. Keywords:*smartphone notifications*, *notification batching*, *interruption*, *phone checking*, *fear of missing out*, *problematic smartphone use*, *notification importance / prioritization*, *gambling marketing*, *dark patterns*; in Turkish*akıllı telefon bağımlılığı*, *gelişmeleri kaçırma korkusu*, *nomofobi*, *üniversite öğrencileri*. Most sources are from 2014–2026; two older foundational studies on interruption and checking habits (Mark et al., 2008; Oulasvirta et al., 2012) were also included.

Selection criteria: A study was included when it met all three conditions: (1) it is a peer-reviewed journal or conference paper; (2) it presents original data (device logs, experiment, survey or interviews) or is a review or meta-analysis; (3) it links directly to a design decision of Süzgeç. Non-peer-reviewed preprints, reports and news were used only for current or Turkey-specific information or to support peer-reviewed findings, and they are flagged in Table 3. Sources whose full text or abstract we could not access, or whose numbers we could not verify, were excluded; for example, the Turkish adaptations of the Bergen Shopping Addiction Scale and the PGSI were left out for this reason.

Table 2. Themes and their links to the research questions

Theme	Question answered	Counterpart in Süzgeç	RQ
1. Definitions, scales and prevalence	How is the problem defined and how is it measured in Turkey?	No diagnosis; filters the areas the user selects; Turkish scales in the pilot	RQ1
2. Interruption and attention cost	Do notifications disrupt attention even when not looked at?	Unimportant notifications silent and off the panel	RQ1
3. Checking and unlocking	What triggers checking, and how often?	Success metric: number of unlocks and checks	RQ1
4. FoMO	How is FoMO related to problematic use?	The “you will hear it if it is Critical” guarantee	RQ2
5. Notification interventions	Turning off, batching or deferring?	Critical and Important immediately, the rest 3 times a day	RQ2
6. Addiction-driving notifications	How should gambling, campaign and social media notifications be filtered?	Content-level filter, Deleted box	RQ3
7. Smart notification management	Can importance be predicted automatically and per person?	Hybrid model, ranking that learns from corrections	RQ3
8. Time limits	How strict should restriction be?	Optional time limit in phase 3	RQ2

1.3 Summary and synthesis

The 52 reviewed studies and sources are summarised in a single table, Table 3. The columns are the fields required by the guidelines: aim, sample and method, main finding, contribution to Süzgeç and limitation. Non-peer-reviewed sources are flagged in the Study column. The table is followed by a short synthesis of each theme and, at the end of the section, the agreeing findings, diverging findings and common limitations across themes.

Table 3. Reviewed studies (Th: theme)

Study	Th	Aim; sample and method	Main finding	Contribution to Süzgeç	Limitation
Panova and Carbonell (2018)	1	Does “smartphone addiction” meet addiction criteria; critical review	Not enough evidence to confirm the addiction framework	The term “problematic use”; Süzgeç does not diagnose	Not systematic; no quantitative synthesis
Akpınar Aslan et al. (2021)	1	Addiction risk among Turkish university students; 612 students, survey	Possible addiction risk in 24.7%; higher at ages 18–24	Rationale for the target users	Cross-sectional; self-report
Noyan et al. (2015)	1	Turkish validity of the Smartphone Addiction Scale – Short Version; 367 students	α = 0.867, test–retest 0.926	Pre–post measure in the pilot	Single university
Gökler et al. (2016)	1	Turkish adaptation of the FoMO Scale; 200 students	α = 0.81; positively related to problematic use	Turkish measure for the goal of “not increasing” FoMO	Small, single university
Demirci (2019)	1	Turkish adaptation of the Bergen Social Media Addiction Scale; factor analyses	Single-factor structure retained; weak positive relation with anxiety and depression	Optional measure for users who choose the social media filter	Self-report; cross-sectional
Yildirim et al. (2016)	1	Prevalence of nomophobia among Turkish students; 537 students, survey	Nomophobia in 42.6% of students	The problem is measured locally in the target group	Cross-sectional
Fournier et al. (2026)	2	Cognitive effect of notifications; 180 students, 3 groups, Stroop, pupillometry, iPhone logs	Slowdown lasting about 7 s; effect largest for personal notifications; related to notification volume and checking frequency, not screen time	Unimportant notifications should not be shown at all	Laboratory; iPhone only

Study	Th	Aim; sample and method	Main finding	Contribution to Süzgeç	Limitation
Stothart et al. (2015)	2	Attention cost of an unanswered notification; sustained attention task, 3 groups	Notifications not looked at still increased errors	Unimportant notifications should make no sound or vibration	Sample from secondary report; laboratory
Kushlev et al. (2016)	2	Effect of alerts on inattention; 221 people, 2 weeks, within-subjects	With alerts on, inattention (d = 0.44) and hyperactivity (d = 0.45) increased	Number of sounds and vibrations is a primary target	Self-report; participants applied the setting themselves
Pielot et al. (2014)	2	Real notification load; 15 people, 1 week of device logs	63.5 notifications a day; viewed within minutes even in silent mode (Figures 1, 3)	Silent mode is not enough; removing from the panel is needed	15 people; 2013 data
Mehrotra, Pejovic et al. (2016)	2	What determines receptivity to notifications; 20 people, 10,372 notifications	Even notifications with important content can be disruptive; the sender relationship affects the decision	Sender and content should be modelled together	Small sample
Mark et al. (2008)	2	Cost of interruptions on work; 48 people, experiment	Work finished with the same quality, but stress, time pressure and effort increased	The cost of interruption shows in stress and fatigue	Laboratory; short task
Shen (2025)	2	Distractions in digital reading; 124 experiments in 32 studies, meta-analysis	Comprehension dropped (g = −0.64)	Academic cost of interruption for students	High heterogeneity (I^2 = 88%); not notification-specific
Eddington et al. (2026), preprint – not peer-reviewed	2	Effect of notification suppression on programming; 22 students, within-subjects	Breaks fell by about 19%; no change or an increase for 8 students	The effect varies from person to person	Not peer-reviewed; small sample
Andrews et al. (2015)	3	Estimated vs. real use; 23 people, 2 weeks of logs	85 uses a day, half shorter than 30 s; use underestimated by half	Device logs are essential; self-report is unreliable	Small sample
Oulasvirta et al. (2012)	3	Structure of the checking habit; usage logs + field experiment	Checks shorter than 30 s; quickly accessible new information builds the habit	Süzgeç's inbox must not become a new checking habit	Early smartphone era
Heitmayer and Lahlou (2021)	3	What triggers an interaction; 37 people, wearable camera	89% started without a notification	Notification management is not enough; checking should also be measured	37 people
Telzer and Burnell (2026)	3	Checking at school and cognitive control; 51 youths aged 15–18, 14 days of iOS logs	64 checks during school hours; frequent checking related to lower cognitive control; screen time unrelated	Number of checks as the main success metric	Cross-sectional; minor sample, supporting evidence only
Lee et al. (2014)	3	Separating problematic use from logs; 95 students, about 27 days	79% of sessions externally triggered; risk group more sensitive to notifications	Session length after a notification should be measured (phase 3)	Broad definition of "external trigger"; single country
Przybylski et al. (2013)	4	Defining and measuring FoMO; 3 studies, 3,179 people	FoMO related to social media use (β = 0.40)	Definition of the FoMO goal	Cross-sectional
Zhang et al. (2023)	4	Meta-analysis of FoMO and addiction; 85 studies	r = 0.47; age, gender and culture did not change the relationship	Süzgeç must not increase FoMO	Review of cross-sectional studies
Rozgonjuk et al. (2019)	4	FoMO, disruption by notifications and learning; 316 students, survey	FoMO related to how often students are disrupted, not to the number of notifications	Hiding unimportant notifications is more valuable for high-FoMO students	Cross-sectional; self-report
Elhai et al. (2016)	4	FoMO, need for touch and anxiety; 308 people, online survey	Problematic use most related to anxiety, need for touch and FoMO	FoMO is a central variable	Cross-sectional; online sample
Gökçearslan et al. (2023)	4	FoMO and addiction in Turkish adults; 488 people, structural equation model	FoMO → addiction 0.50 (usage time 0.22)	FoMO is central in Turkey as well	Cross-sectional; convenience sample
Fitz et al. (2019)	5	Batching; RCT, 237 people, 2 weeks, 4 conditions	Delivery 3 times a day reduced unlocks (d = −0.60); no notifications increased FoMO and anxiety	Source of the 3-times-a-day delivery idea	Notifications batched without distinction; 2 weeks
Dekker et al. (2025)	5	Turning off all notifications; pre-registered RCT, 205 young adults, 1 week	Notifications fell from 154 to 8, checking did not change, FoMO increased (d = 0.89)	Filtering, not turning off	1 week; Android only
Pielot and Rello (2017)	5	24 hours without notifications; 30 people, within-subjects	More productive but anxious about delayed replies	Critical must get through	24 hours; volunteer sample

Study	Th	Aim; sample and method	Main finding	Contribution to Süzgeç	Limitation
Castelo et al. (2025)	5	Blocking mobile internet; RCT, 467 people, 2 weeks	Well-being improved, but only 25.5% kept the block	A selective approach instead of hard restriction	Low compliance; iPhone only
Kushlev and Dunn (2015)	5	Checking e-mail 3 times a day; 124 people, within-subjects	Stress decreased (d = 0.37)	Second piece of evidence for batching	Self-report
Okoshi et al. (2017)	5	Deferring notifications to an opportune moment; more than 680,000 users, 3 weeks	Response time 49.7% shorter; engagement increased	Deferring does not lose users	Notifications of a single app
Olson et al. (2022)	5	Nudge-based intervention; two studies with 51 and 70 people, 2–6 weeks	Screen time and problematic use decreased; the effect lasted 6 weeks	Turning off unnecessary notifications is part of effective packages	Separate effects of components cannot be measured
Rockloff et al. (2026)	6	Direct gambling marketing; randomised field experiment, 227 bettors, 14 days	Those opted out of marketing spent 39% less and reported 67% less harm	Gambling goes to Deleted by default	Australia; self-report
National Consumers League (2025), report – not peer-reviewed	6	3 major betting apps, 4 weeks, 108 notifications	93% advertising; no option to turn off advertising only	A content-level filter is needed	Not peer-reviewed; small sample
Bouguettaya et al. (2020)	6	Gambling advertising and behaviour; critical and meta-analytic review	Exposure related to attitudes, intentions and behaviour	The less advertising is seen, the better	Few experimental studies
Perdana et al. (2024)	6	Detecting gambling promotion from text; 6,038 tweets	TF-IDF + Random Forest F1 0.96	Can be separated from short text	Different language and platform
Sathvik et al. (2026), preprint – not peer-reviewed	6	Dataset of manipulative betting ads; 3,779 ads	Deceptive class hardest (F1 0.43–0.51)	Source for a gambling lexicon	Dataset not yet released
Monge Roffarello, Lukoff and De Russis (2023)	6	Attention-capturing deceptive designs; review of 43 papers	11 patterns; one is “fake social notifications”	Filtering non-message social media notifications	Conceptual
Temel (2024)	6	Shopping notifications and impulsive buying; 17 people, interviews	Those who respond positively to notifications tend toward impulsive buying	Shopping: delete campaigns only	Qualitative; not generalisable
Yeşilay (2026), news – not peer-reviewed	6	Gambling prevalence and applications in Turkey	Applications: 37 in 2019, 5,748 in 2025; 42% of clients aged 18–25	Turkish context	Reported via news
Sahami Shirazi et al. (2014)	7	Subjective value of notifications; 40,191 users, about 200 million notifications	Messaging most important (4.43/5), system least important (1.69)	Category-based starting rules (Figures 8–9)	Self-selected users; no content
Visuri et al. (2019)	7	Importance prediction; 40 users, 113,197 notifications, on-device model	91.1% accuracy; 44.2% of swiped-away notifications were important	Closest predecessor (Figures 10–12)	40 people
Lin, T.-C. et al. (2021)	7	Desired notification order; 34 people, 4 weeks, 18,439 ranked notifications	Order related to importance, urgency and sender; default order does not fit	Importance-ordered inbox	Single country
Lin, Y.-H. et al. (2024)	7	Sorting and categorising; 30 people, 21 days, 72,497 notifications	Generic automatic sorting preferred only 13.77% of the time; pinning most valuable	Learning from corrections (Figure 13)	The system did not learn
Mehrotra, Hendley and Musolesi (2016)	7	Turning preferences into understandable rules; 18 people, 11,185 notifications	57% of suggested rules were accepted	Showing learned rules is feasible	Small sample
Katevas et al. (2017), preprint – not peer-reviewed	7	Will a notification be attended to; 279 people, 446,268 notifications	XGBoost and LSTM similar (AUC about 0.70)	Classical model sufficient for numeric context	Not peer-reviewed
Zheng et al. (2025), preprint – not peer-reviewed	7	Personalised urgency; 18 people, large language model	Accuracy rose from 67% to 81.5%	Personalisation is needed	Not peer-reviewed; does not run on the phone
Chen et al. (2025)	7	Notification summaries with a large language model; 79 users + 20 interviews	Summaries lost context; 72% quit within 3 days	Süzgeç gives counts, not summaries	Cloud model; high drop-out

Study	Th	Aim; sample and method	Main finding	Contribution to Süzgeç	Limitation
Heinisch et al. (2022), preprint – not peer-reviewed	7	Mood and response time; 18 people, 5 weeks	Response time varies greatly by person and app	Rationale for personalisation	Not peer-reviewed; small sample
Kim et al. (2019)	8	Lockout mechanisms; 36 people, 4 weeks, 3 conditions	Strong lockout most effective but most tiring; 52.8% chose the weak lockout	Mandatory lock is optional	1 week per condition
Grüning et al. (2023)	8	Short delay at app launch; 280 people, 6 weeks	Target app openings fell by 57%	Short delay on the limit-exceeded screen	Long-term effect unknown
Allcott et al. (2022)	8	Self-control in digital addiction; 1,933 people, 12-week RCT	Self-set limits reduced use by 16%	The user sets the limit	USA; Android only
Monge Roffarello and De Russis (2023)	8	Self-control tools; systematic review of 62 papers	Small-to-medium effect; evaluations short-term	Comparison and follow-up in the pilot	Limited quality of reviewed studies

Theme 1 – Definitions, scales and prevalence

Gambling disorder is an official diagnosis (American Psychiatric Association, 2013; World Health Organization, 2019), but there is no separate diagnostic category for smartphones, social media or shopping, and whether the concept counts as an addiction is debated (Panova & Carbonell, 2018). Süzgeç therefore does not diagnose; it filters the areas the user chooses. In Turkey, about a quarter of students show addiction risk (Akpınar Aslan et al., 2021) and 42.6% show nomophobia (Yildirim et al., 2016). Three scales with shown Turkish validity exist (Noyan et al., 2015; Gökler et al., 2016; Demirci, 2019), but all rely on self-report. The primary success metric will therefore be logged behaviour; the scales can be used only for pre–post comparison in the pilot.

Theme 2 – Interruption and attention cost

Eight studies point in the same direction: a notification breaks attention even when the user does not touch it (Stothart et al., 2015; Fournier et al., 2026), and this cost accumulates as stress and inattention (Mark et al., 2008; Kushlev et al., 2016). The cost grows with how important the notification seems to the person, and even important content can be disruptive (Mehrotra, Pejovic et al., 2016). There is an academic cost for students as well (Shen, 2025). In Pielot et al. (2014), notifications were viewed within minutes even when the phone was on silent (Figure 3); this explains why Süzgeç does not stop at silencing but also removes the notification from the panel. The effect of suppression varies from person to person (Eddington et al., 2026), which links to personalisation.

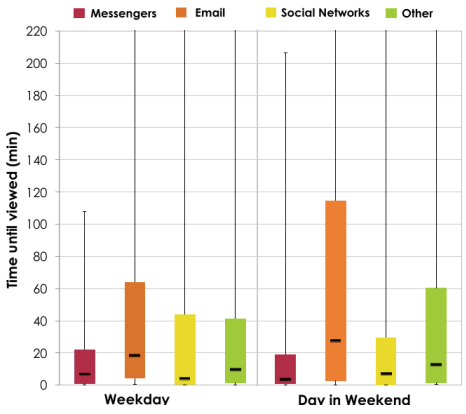


Figure 3. Median time between receiving and viewing a notification (minutes), by notification type, weekdays and weekends. Source: Pielot et al. (2014), Figure 4.

Theme 3 – Checking and unlocking

Checking is frequent (about 85 times a day; Andrews et al., 2015; Dekker et al., 2025), short (under 30 seconds; Oulasvirta et al., 2012) and often starts without a notification (Heitmayer & Lahlou, 2021). Checking frequency is more consistently related to cognitive cost than screen time is (Telzer & Burnell, 2026; Fournier et al., 2026). Because quickly accessible new information feeds the checking habit (Oulasvirta et al., 2012), Süzgeç's inbox could also become a new habit; we will therefore count inbox visits outside delivery times. The difference between Heitmayer and Lahlou (89% user-initiated) and Lee et al. (79% externally triggered) is discussed in Table 5.

Theme 4 – FoMO

FoMO is consistently related to problematic use (Przybylski et al., 2013; Elhai et al., 2016; Zhang et al., 2023) and is the strongest variable in the Turkish sample as well (Gökçearslan et al., 2023). The finding of Rozgonjuk et al. (2019) matters for Süzgeç: students with high FoMO do not receive more notifications, but they are disrupted more by the same notifications. In Fitz et al. (2019), cutting off notifications entirely increased anxiety through phone FoMO (Figure 4). Süzgeç therefore gives the guarantee "what is important arrives now, the rest at set times" rather than "nothing will come". The studies in this theme are cross-sectional; we use FoMO as a variable to monitor, not as a causal outcome.

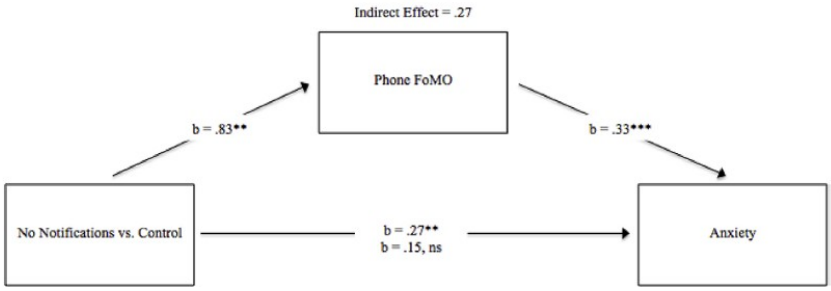


Figure 4. Effect of receiving no notifications on anxiety through phone FoMO (indirect effect 0.27). Source: Fitz et al. (2019), Fig. 4.

Theme 5 – Notification interventions: turning off, batching, deferring

The idea of Süzgeç came from Fitz et al. (2019): in a randomised field experiment with 237 people, batched delivery 3 times a day (09:00, 15:00 and 21:00; Figure 5) led to fewer unlocks (d = −0.60), a greater sense of control (d = 0.58), less inattention (d = −0.65) and higher productivity (d = 0.57) than the control group; hourly delivery did not bring the same benefit (Figure 6). Among those receiving no notifications, FoMO (d = 0.59) and anxiety (d = 0.56) increased. Even with delivery 3 times a day, fear of missing notifications increased (d = 0.68), and the authors suggested personalisation, noting that "not all notifications are equally important"; Süzgeç letting Critical and Important through without waiting answers both points.

Other studies complete the picture. Turning everything off reduces the number of notifications but does not reduce checking and increases anxiety (Pielot & Rello, 2017; Dekker et al., 2025). Predictable batching (Kushlev & Dunn, 2015) and deferring to an opportune moment (Okoshi et al., 2017) reduce interruption while keeping the benefit of notifications. According to the data of Visuri et al. (2019), engagement with a new notification falls when many notifications arrive in a short time (Figure 7); a constant stream devalues notifications. Turning off unnecessary notifications is part of effective intervention packages (Olson et al., 2022), but hard restriction is hard to sustain (Castelo et al., 2025). Süzgeç combines these findings: Critical and Important immediately, the rest 3 times a day, restrictions under the user's control.

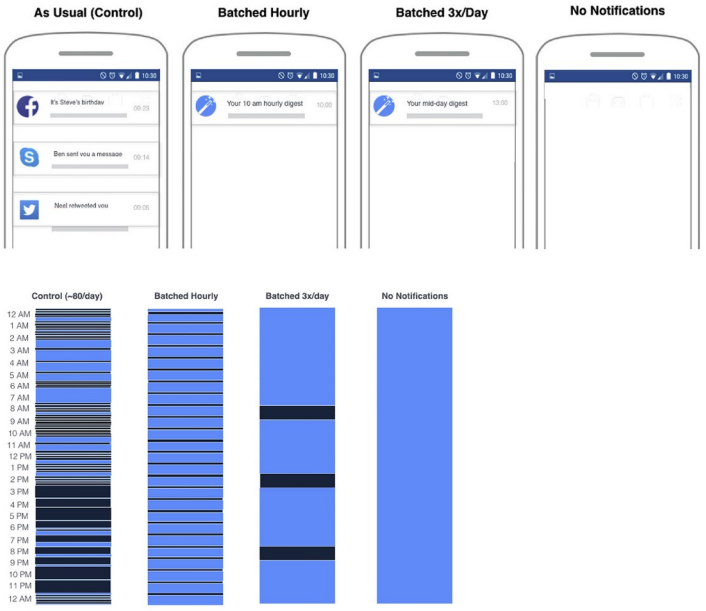


Fig. 1. Examples of participants' notification delivery in each condition.

Figure 5. Example notification deliveries by condition (as usual, hourly, 3 times a day, no notifications). Source: Fitz et al. (2019), Fig. 1.

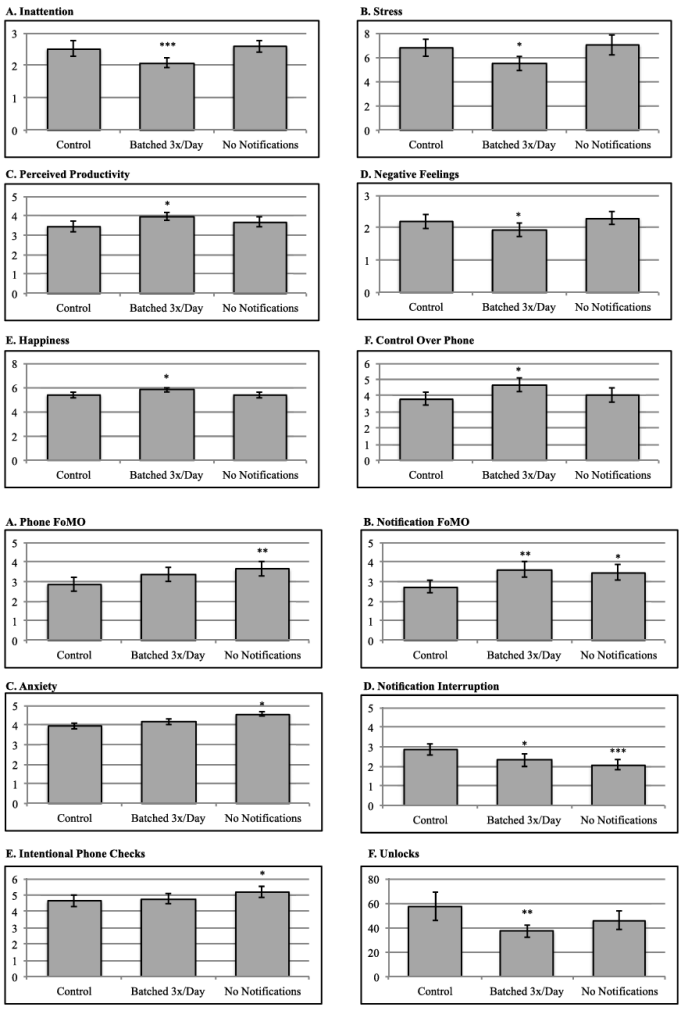


Fig. 2. *The effects of Batching 3x/Day and No Notifications on main effects. Note. Error bars represent 95% CIs, *p < .05, **p < .01, ***p < 001.*

Figure 6. Effects of batched delivery 3 times a day and of receiving no notifications on the main outcomes (95% CI). Source: Fitz et al. (2019), Fig. 2.

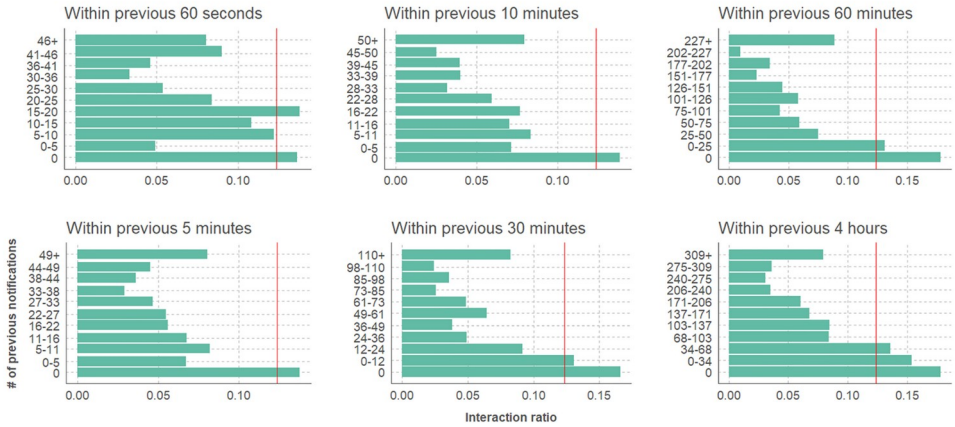


Figure 7. Likelihood of interacting with a new notification by the number of notifications that arrived earlier within a given time window; the red line marks the mean interaction rate (12.3%). Source: Visuri et al. (2019), Fig. 5.

Theme 6 – Addiction-driving notifications

The sources show two things together: gambling marketing causes harm (Bouguettaya et al., 2020; Rockloff et al., 2026), and betting apps do not let users separate advertising from security alerts such as logins and withdrawals (National Consumers League, 2025). The filter must therefore work at the content level, not the app level. Perdana et al. (2024) and Sathvik et al. (2026) show that this can be done from short text; however, deceptive language is the hardest class. Fake social notifications on social media (Monge Roffarello, Lukoff & De Russis, 2023) and campaign notifications in shopping (Temel, 2024) are other examples of the same logic.

Turkish context: Applications to Yeşilay because of gambling rose from 37 in 2019 to 5,748 in 2025 (Yeşilay, 2026). The Information and Communication Technologies Authority (BTK) decided that SMS messages with links from abroad will not be delivered from 1 April 2026 (NTV, 2026); Law No. 6563 makes commercial electronic messages subject to the recipient's consent, which can be withdrawn through the Message Management System (İYS). These measures are limited to SMS and commercial messages; app notifications and campaigns the user once agreed to keep arriving. Süzgeç closes this gap on the user's side. Yeşilay's free YEDAM 115 line provides counselling for gambling and technology addiction.

Theme 7 – Smart notification management and priority classification

Nine studies lead to three conclusions. (1) Importance can be predicted: app category (Sahami Shirazi et al., 2014; Figures 8–9), sender and content (Lin, T.-C. et al., 2021; Mehrotra, Pejovic et al., 2016) are strong signals; the personal model of Visuri et al. (2019) reached 91.1% accuracy, and the type of interaction varies strongly by category (Figure 10). (2) Non-personal ranking is not enough: in Lin, Y.-H. et al. (2024) generic sorting was preferred only 13.77% of the time; personalisation raised accuracy from 67% to 81.5% (Zheng et al., 2025); response behaviour varies greatly from person to person (Heinisch et al., 2022). Showing learned rules to the user is well accepted (Mehrotra, Hendley & Musolesi, 2016). (3) Implicit behaviour is misleading: a large share of swiped-away notifications are actually important (Visuri et al., 2019). Süzgeç will therefore learn from the user's explicit corrections, not from swipes.

The show/defer/hide table of Visuri et al. (2019) (Figure 11) maps onto Süzgeç's categories: what is important is shown (Critical, Important), what can wait is deferred (Normal, to delivery time), and what is unimportant is hidden (Unimportant, Deleted). The on-device prediction flow of the same study (Figure 12) is the predecessor of the Süzgeç architecture (Figure 19). For numeric context features, classical machine learning performs as well as deep networks (Katevas et al., 2017). In Chen et al. (2025),

summaries losing context and the high drop-out rate support Süzgeç giving only counts, not content summaries, in its delivery notification.

	Number of			
Category	**Users**	**Apps**	**Click times**	**Ratings**
messenger	29,627	10	2,508,203	883
voice & messenger	27,768	9	335,530	470
mail	26,120	8	781,502	610
social	22,173	12	501,159	382
calendar	12,292	5	18,990	305
alarm clock	11,849	7	18,678	50
music	15,366	16	47,786	114
game	3,046	6	43,206	41
market	29,326	6	192,357	156
reader/news	3,958	7	11,036	50
utility	13,202	27	29,990	105
tool	11,494	18	33,352	42
system	27,269	21	229,067	304
other	13,511	21	44,370	124
overall	37,233	173	4,795,226	3,636

Figure 8. The 14 app categories derived by Sahami Shirazi et al. (2014) and the number of users, apps, clicks and ratings in each category. Source: Sahami Shirazi et al. (2014), Table 1.

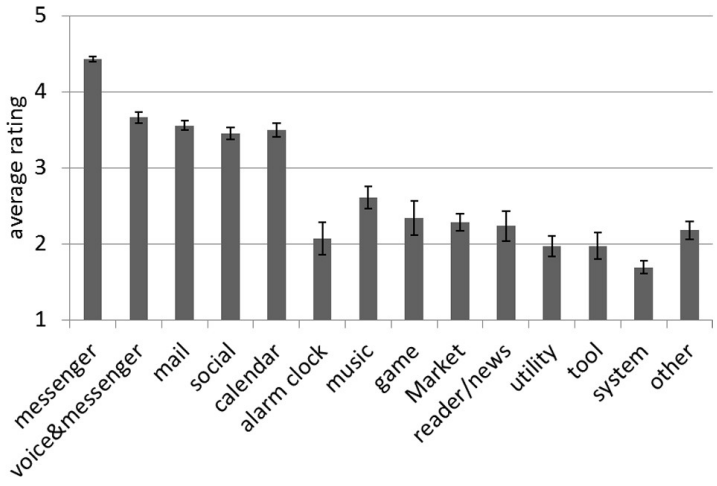


Figure 9. Mean notification importance rating by app category (1–5). Source: Sahami Shirazi et al. (2014), Figure 7.

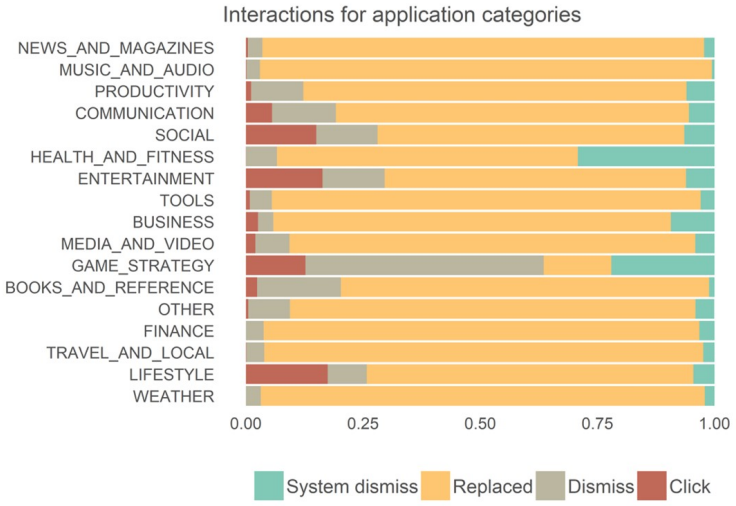


Figure 10. Types of interaction with notifications by Play Store category (click, swipe, replaced by a newer one, removed by the system); categories ordered by mean content importance. Source: Visuri et al. (2019), Fig. 3.

Table 3
Proposed actions for notification filtering based on user-reported information.

		Was the notification appropriately timed?	
		Yes	No
Are the contents important or relevant?	Yes	Always show the notification	Defer until next use
	No	Preferably hide[a]	Always hide the notification

Figure 11. Show/defer/hide actions by content importance and timing suitability. Source: Visuri et al. (2019), Table 3.

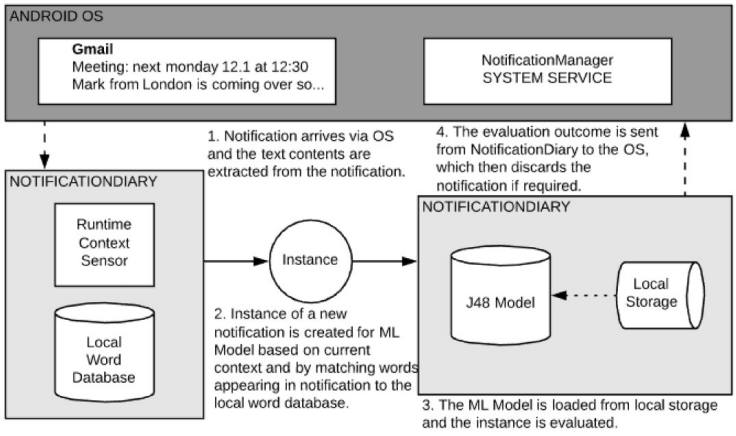


Figure 12. On-device prediction flow of the NotificationDiary app. Source: Visuri et al. (2019), Fig. 7.



Figure 13. NotiManager interface: (a) category menu, (b) automatic sorting indicator, (c) clear all, (d–e) pinning, (f) manual sorting. Source: Lin, Y.-H. et al. (2024), Figure 1.

Theme 8 – Time limits

This theme supports Süzgeç's phase-3 time-limit feature and is therefore kept short. Restriction works, but the stricter it is, the more tiring it becomes (Kim et al., 2019); a short delay (Grüning et al., 2023) and a self-set limit (Allcott et al., 2022) reduce use in less coercive ways. Süzgeç's time limit will therefore depend on the user's choice (mandatory or warning only), and the mandatory lock can be switched off instantly in the settings. Because evaluations in the field are generally short-term (Monge Roffarello & De Russis, 2023), we should not claim lasting effects from our own pilot.

Synthesis across themes

Agreeing findings: Studies using different methods point in the same direction on seven points (Table 4).

Table 4. Agreeing findings

Finding	Supporting studies	Counterpart in Süzgeç
Notifications break attention even when not looked at; the cost grows with perceived importance	Stothart et al. (2015); Kushlev et al. (2016); Mehrotra, Pejovic et al. (2016); Fournier et al. (2026)	Unimportant notifications silent and off the panel
The problem is checking frequency rather than screen time	Andrews et al. (2015); Heitmayer and Lahlou (2021); Fournier et al. (2026); Telzer and Burnell (2026)	Success metric: number of unlocks and checks
Turning everything off increases anxiety; predictable batching works	Kushlev and Dunn (2015); Pielot and Rello (2017); Fitz et al. (2019); Dekker et al. (2025)	Critical and Important immediately, the rest 3 times a day
FoMO is consistently related to problematic use	Przybylski et al. (2013); Elhai et al. (2016); Zhang et al. (2023); Gökçearslan et al. (2023)	Goal of not increasing FoMO
Importance depends on category, sender and content and varies between people	Sahami Shirazi et al. (2014); Lin, T.-C. et al. (2021); Heinisch et al. (2022); Zheng et al. (2025)	Rule layer + personalised model
User control and correctability are essential	Mehrotra, Hendley and Musolesi (2016); Kim et al. (2019); Lin, Y.-H. et al. (2024)	Manual moving and sorting, restorable Deleted box
Addiction-driving marketing causes harm and should be filtered at the content level	Bouguettaya et al. (2020); Perdana et al. (2024); National Consumers League (2025); Rockloff et al. (2026)	Gambling goes to Deleted by default

Diverging findings: On six points studies reach different conclusions; most differences can be explained by methodological differences such as definitions, measurement or how coercive the intervention was (Table 5).

Table 5. Diverging findings and possible explanations

Topic	On one side	On the other side	Possible explanation → conclusion for Süzgeç
Who starts the interaction?	89% the user (Heitmayer & Lahlou, 2021)	79% external trigger (Lee et al., 2014)	Definitions differ → both notifications and checking are tracked
Does turning off improve attention?	Yes (Kushlev et al., 2016)	No, anxiety increased (Fitz et al., 2019; Dekker et al., 2025)	In Kushlev the setting could be undone → control stays with the user
Does hard restriction work?	Attention and well-being improved (Castelo et al., 2025); strong lockout most effective (Kim et al., 2019)	Only 25.5% kept it; 52.8% chose the weak lockout	There is an effect, but low sustainability → light and selective intervention
Does automatic sorting work?	91.1% accuracy (Visuri et al., 2019)	13.77% preference (Lin, Y.-H. et al., 2024)	Personal vs. generic model → learn from corrections
Does swiping mean “unimportant”?	Studies counting clicks as acceptance (Sahami Shirazi et al., 2014)	44.2% of swiped-away notifications important (Visuri et al., 2019)	A swipe can also mean “not now” → learn from explicit corrections
Is deep learning required for text?	Lin, Y.-H. et al. (2024) used BERT	Classical methods suffice (Katevas et al., 2017; Perdana et al., 2024)	Classical methods are strong on short promotional text → hybrid; tested in Part 2

Common limitations: The weaknesses of the reviewed studies are also largely shared (Table 6). We therefore base design decisions not on a single study but on the agreeing findings in Table 4.

Table 6. Common limitations

Limitation	Seen in	Süzgeç’s response
Small samples (15–40 people)	Pielot et al. (2014); Andrews et al. (2015); Visuri et al. (2019); Heitmayer and Lahlou (2021); Lin, T.-C. et al. (2021); Lin, Y.-H. et al. (2024); Zheng et al. (2025)	Base decisions on agreeing findings; collect our own data in the pilot
Short duration (24 hours–6 weeks)	Pielot and Rello (2017); Fitz et al. (2019); Grüning et al. (2023); Castelo et al. (2025); Dekker et al. (2025); Monge Roffarello and De Russis (2023)	No claims of long-term effects; weekly follow-up in the pilot
Self-report measures	Kushlev and Dunn (2015); Kushlev et al. (2016); Rozgonjuk et al. (2019); Gökçearslan et al. (2023); scales in Theme 1	Primary metric from device logs
Cross-sectional design	All of Theme 4; Telzer and Burnell (2026)	Use FoMO as a monitored variable
Laboratory or non-notification-specific measures	Mark et al. (2008); Stothart et al. (2015); Shen (2025); Fournier et al. (2026)	Complement with real notification logs (Part 2)
No notification data from Turkey	From Turkey only scales, surveys and a 17-person qualitative study (Temel, 2024)	Turkish text data and synthetic notifications (Part 2)
Non-peer-reviewed sources	Katevas et al. (2017); Heinisch et al. (2022); Zheng et al. (2025); National Consumers League (2025); Eddington et al. (2026); Sathvik et al. (2026); Yeşilay (2026)	Flagged in the table; no decision relies on them alone

1.4 Gaps and conclusion

The reviewed sources support Süzgeç's decisions one by one; however, we found no study that tests them together (Table 7).

Table 7. Research gaps and Süzgeç's response

Gap	Basis	Süzgeç's response
G1 – Importance-aware batching has not been tested	Fitz et al. (2019) batched without distinction; Visuri et al. (2019) did not implement deferral	Critical and Important immediately, the rest 3 times a day
G2 – No tool filters addiction content at the notification level	National Consumers League (2025); classification studies are on social media	Content filter based on the selected area
G3 – No notification inbox that learns from corrections	In Lin, Y.-H. et al. (2024) the system did not learn	Ranking that learns from corrections
G4 – No Turkish notification data	Studies in Turkey are surveys and scales	Turkish SMS, e-mail and synthetic data (Part 2)
G5 – Missing important notifications has not been measured objectively	Only self-report in Fitz et al. (2019)	Missed Critical notifications are measured

Contribution: Süzgeç was inspired by the experiment of Fitz et al. (2019) and starts from the authors' limitation: not all notifications are equally important. This is not a claim to be "the first"; it is based on the scope of the sources we reviewed.

From literature to metrics: Because screen time is unrelated to cognitive cost (Fournier et al., 2026), it is not a primary metric. The success metrics are in Table 8. The demo and mini-pilot are a feasibility demonstration; we will report the results as trends.

Table 8. Success metrics

Metric	Basis	How it will be measured
H1 – Not missing Critical notifications	Fitz et al. (2019); Dekker et al. (2025)	Critical recall on the test set; number of missed Critical notifications in the pilot
H2 – Classification performance	Visuri et al. (2019); Perdana et al. (2024)	Macro F1 on the test set
H3 – Number of sound or vibration alerts	Stothart et al. (2015); Kushlev et al. (2016)	Logging mode compared with active mode
H4 – Number of unlocks and checks	Fitz et al. (2019); Telzer and Burnell (2026)	Digital Wellbeing data; logging mode vs. active mode
H5 – Catching addiction content	Rockloff et al. (2026); Perdana et al. (2024)	Gambling and campaign recall on the test set
H6 – User corrections	Lin, Y.-H. et al. (2024)	Weekly correction rate (expected to fall)
The inbox not becoming a habit	Oulasvirta et al. (2012)	Inbox visits outside delivery times

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Part 2 – Data Review

2.1 Introduction

The data review seeks answers to five questions:

- **RQ-D1:** What share of notifications is persistent or short-lived; how much do Android flags and the 5-second rule filter out?
- **RQ-D2:** How many notifications does a user receive per day, and from which apps?
- **RQ-D3:** How much does the notification profile vary between people?
- **RQ-D4:** Can Turkish campaign messages be separated from normal SMS and e-mails by their text?
- **RQ-D5:** How well does a simple text model make this separation, and how well does it generalise to another dataset?

Exploration is needed because, if Süzgeç's rules (flags, the 5-second wait, delivery 3 times a day) are not validated with data, either the inbox fills with unnecessary notifications or Critical notifications are missed. In addition, the datasets we use were collected for other purposes and contain duplicates and logging errors.

2.2 Structure

Section 2.3 covers the datasets, 2.4 ethics, 2.5 the analysis of notification logs, 2.6 Turkish text analysis and a classification experiment, 2.7 data quality and 2.8 the conclusion. Analyses were done in Python (pandas, scikit-learn; Pedregosa et al., 2011); the random seed for steps involving randomness is 42. All figures in this section (Figures 14–18) were not taken from papers; they were produced by our team from the raw data.

2.3 Data description

Table 9. Datasets evaluated

Dataset	Content and size	Role	Licence
Komninos et al. (2019)	44 Android users, 235,423 notification records (December 2018–May 2019); app, flags, time, location; no text	Main analysis (RQ-D1–D3)	GPL-3.0, attribution
Komninos et al. (2018)	14 users, 176,195 notifications	Backup and validation	GPL-3.0, attribution
TurkishSMSCollection (Karasoy and Ballı, 2021)	4,751 SMS (2,536 spam, 2,215 normal)	Turkish text analysis and model training (RQ-D4, D5)	Not specified; research use only
Uysal et al. (2013)	850 SMS (420 spam, 430 normal)	External test on another SMS set (RQ-D5)	Not specified; attribution requested
UCI Turkish Spam V01	826 e-mails (330 spam, 496 normal)	E-mail analysis and testing (RQ-D4, D5)	CC BY 4.0
LSApp (Aliannejadi et al., 2021)	292 users, 599,635 app usage records	Phase 3 (transition analysis); not analysed in this assignment	No licence file

Evaluated but not used: BetXplain (data not yet released) and a Turkish SMS set with restricted access. In addition, our team will collect its own real notifications and generate synthetic Turkish notifications (including gambling).

2.4 Ethics

Secondary data: Identities are anonymous; we did not use the location columns in Komninos (2019) (place name, latitude–longitude); the data are not redistributed; datasets with unclear licences are used for research only. We did not quote sample messages from the e-mail and SMS texts; we report only aggregate statistics.

The Süzgeç app: Notification texts may contain third parties' messages, one-time codes and banking information. Therefore all classification will run on the device and no text will be sent to a server. We plan to follow the KVKK (Turkish Personal Data Protection Law) principles of purpose limitation, data minimisation, explicit consent and the right to erasure; a detailed compliance study will be done in the final phase. The app is for users aged 18+.

2.5 Analysis of notification logs

Cleaning (Komninos 2019): 1 record outside the study period (dated 2015) was removed. 40,329 records (17%) that were updates of the same notification (same user, app, notification ID and posting time) were merged into a single notification, leaving 195,093 notifications. Per-person daily analyses used the 37 users with at least 7 days of data.

Finding 1 – The rule layer handles half of the load (RQ-D1): 49.6% of notifications carry a persistent or service flag; they can be handled in the rule layer without reaching the model. 26% of the remaining notifications disappear by themselves within 5 seconds (Figure 14b; 10–39% for most users). The 5-second wait filters these notifications out without ever adding them to the inbox.

Finding 2 – The load is high and concentrated in one app (RQ-D2): The median number of non-persistent notifications per person per day is 86 (Figure 14a; interquartile range 55–116; 95% CI 65–106, bootstrap), consistent with the 63–154 range in Section 1.3. With delivery 3 times a day, each delivery will contain about 25–30 notifications, so importance ranking is needed in the inbox. 46% of non-persistent notifications come from Facebook Messenger, followed by Instagram (7%), Play Store (4%), Viber (4%) and Gmail (3%) (Figure 15a). Because the load is concentrated in a few apps, app-level starting rules (e.g. Play Store updates as Unimportant) can handle a large share of it on their own.

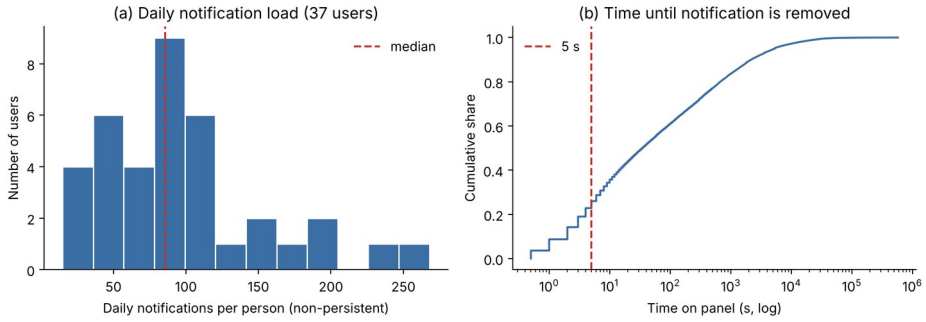


Figure 14. Our own analysis of the Komninos et al. (2019) data: (a) distribution of daily non-persistent notifications per person (37 users with at least 7 days of data); (b) cumulative distribution of how long non-persistent notifications stay on the panel (dashed line: 5 seconds).

Finding 3 – The profile varies strongly between people (RQ-D3): The median share of messaging apps per person is 55%, but it is below 11% for 10% of users and above 87% for another 10% (Figure 15b). For one user most notifications are messages; for another, most are app and service notifications; a single rule set does not fit everyone. This supports the personalisation finding in the literature (Table 4) and is the rationale for Süzgeç's ranking that learns from corrections.

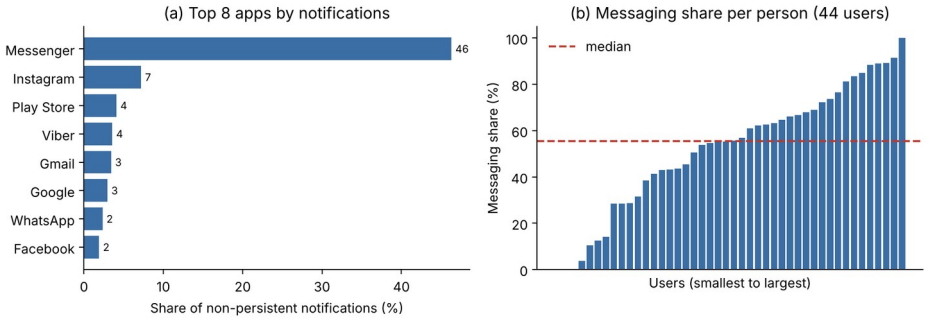


Figure 15. Our own analysis of the Komninos et al. (2019) data: (a) share of the 8 apps sending the most non-persistent notifications; (b) share of messaging apps (Messenger, WhatsApp, Viber, Telegram, SMS, etc.) for each of the 44 users, sorted from smallest to largest.

2.6 Turkish text analysis and classification experiment

Finding 4 – Campaign language is distinct in SMS (RQ-D4): Spam SMS have a median of 21 words, normal messages 6 words (Figure 16). 55% of spam messages contain a link and 82% contain a number with three or more digits; among normal messages this share is below 1%. The most distinctive words are “indirim” (discount), “ücretsiz” (free), “hediye” (gift), “fırsat” (deal) and, from the İYS opt-out text, “ret” (opt-out), “istemiyorsanız” (if you do not want) and “mersis” (company registry number). Campaign notifications can be separated from short text; however, the set has no gambling label.

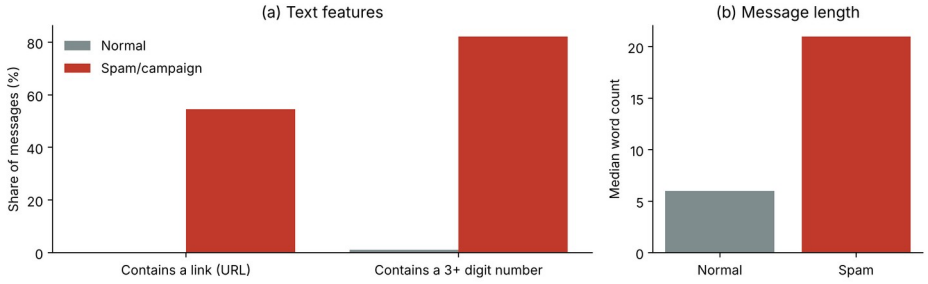


Figure 16. Our own analysis of the TurkishSMSCollection data (4,751 SMS): (a) share of messages containing a link and a number with three or more digits; (b) median word count.

Finding 5 – Simple rules do not work for e-mail (RQ-D4): The picture is different in the e-mail set: 70% of normal e-mails also contain a link and 72% a number with three or more digits (Figure 17); spam e-mails are longer (median 223 vs. 97 words). The most distinctive words come from bulk promotions targeting institutions and graduates (“mezunları” (graduates), “kpss” (civil service exam), “atama” (appointment), “kurumunuzun” (of your institution)), while those of normal e-mails come from everyday correspondence (“evet” (yes), “şimdi” (now), “biraz” (a little)). In other words, the rule “if it has a link or a number, it is a campaign” that works for SMS breaks down for e-mail; a content model is needed.

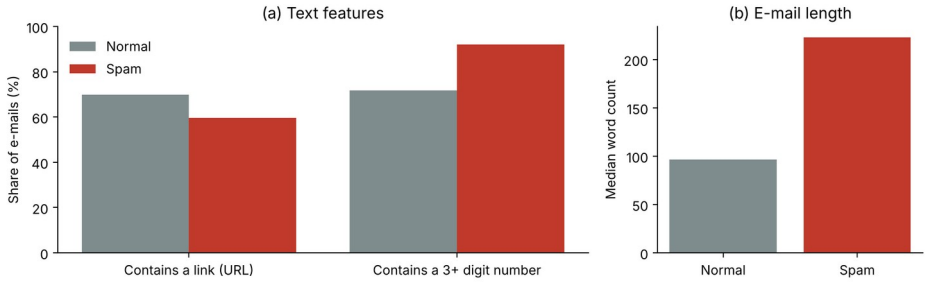


Figure 17. Our own analysis of the UCI Turkish Spam V01 data (825 e-mails analysed): (a) share of e-mails containing a link and a number with three or more digits; (b) median word count.

Finding 6 – The simple model is strong but does not generalise to another text type (RQ-D5): We tested TF-IDF + logistic regression, proposed as the baseline model in Part 3, in four set-ups: 5-fold stratified cross-validation within the same data, training on SMS and testing on another SMS set (Uysal et al., 2013), training on SMS and testing on e-mail, and training on e-mail and testing on SMS. As a comparison we used the rule "spam if it contains a link or a number with 3+ digits" (Table 10, Figure 18).

Table 10. Classification experiment results (precision and recall for the spam class)

Set-up	Method	Accuracy	Macro F1	Precision	Recall
SMS, 5-fold CV	Rule	0.962	0.962	0.990	0.938
SMS, 5-fold CV	TF-IDF (word) + LR	0.992	0.992	0.992	0.993
SMS, 5-fold CV	TF-IDF (character) + LR	0.998	0.998	0.999	0.997
E-mail, 5-fold CV	Rule	0.478	0.428	0.431	0.967
E-mail, 5-fold CV	TF-IDF (word) + LR	0.941	0.937	0.967	0.881
E-mail, 5-fold CV	TF-IDF (character) + LR	0.945	0.943	0.949	0.912
Uysal SMS (no training)	Rule	0.835	0.834	0.887	0.764
SMS → Uysal SMS	TF-IDF (character) + LR	0.953	0.953	0.913	1.000
SMS → E-mail	TF-IDF (character) + LR	0.618	0.617	0.513	0.848
E-mail → SMS	TF-IDF (character) + LR	0.569	0.504	1.000	0.193

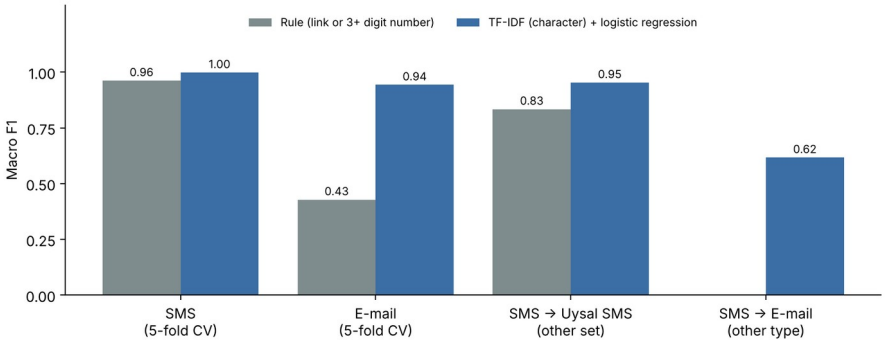


Figure 18. Macro F1 of the rule and of the TF-IDF (character) + logistic regression model: within the same data, on another SMS set and on another text type (our own experiment).

Three conclusions emerge. (1) On the same type of data the simple model is almost perfect (macro F1 0.998 on SMS), consistent with the result of Perdana et al. (2024) on gambling tweets (F1 0.96). (2) On another SMS set performance drops but stays high (0.953) and is clearly better than the rule (0.834). (3) When the text type changes, the model breaks down: the model trained on SMS falls to 0.617 on e-mail, and the model trained on e-mail falls to 0.504 on SMS. Notification texts are a third type that resembles neither SMS nor e-mail; retraining the model on team data and synthetic Turkish notifications is therefore essential. The model is also small: the word-based version is about 0.45 MB, the character-based version about 1.4 MB, and prediction takes about 0.5 ms per message (measured on a laptop). This supports the "TF-IDF + LR as fallback model" decision in Part 3. Limitation: the labels in this experiment are spam/normal; Süzgeç's five-category importance classification and the gambling label have not been tested yet.

2.7 Data quality

Table 11. Data quality issues

Issue	Evidence	Solution
Update duplicates	17% of records are updates of the same notification	Merged; in the app, updates do not create a new row
Difference from the published count	235,423 records in the file, 229,792 in the published description	The difference is reported; analysis uses the cleaned data
No notification text	Komninos sets contain only app and flags	The text model is trained on Turkish SMS, e-mail and synthetic data
Malformed file	Unquoted texts with commas and line breaks in the e-mail set	Taking the last field as the label recovered 825 of 826 rows (1 row without a label)
No gambling label	SMS and e-mail sets have only spam and normal	Synthetic gambling notifications
Difference between text types	The model trained on SMS breaks down on e-mail (Table 10)	Retraining on notification text
Old and foreign data	2018–2019, Greece	Team data and synthetic Turkish data
Rare Critical class	Few Critical notifications	Class weighting; success metric is Critical recall

2.8 Conclusion

The data support Süzgeç's hybrid design: half of the notifications are filtered by flags and a quarter of the rest by the 5-second rule without reaching the model. Because the daily load is high and concentrated in a few apps, both app-level starting rules and importance ranking are needed; because the profile varies by person, personalisation is essential. Turkish campaign language in SMS can be separated even with simple rules, but the rules break down in e-mail, and the model also breaks down when the text type changes. Behind the rule layer, therefore, there will be a content model trained on real notification texts and synthetic Turkish notifications (including gambling).

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- For Perdana et al. (2024), see 1.5.

Part 3 – Technology Review

3.1 Introduction

This section gives the technology choice and its rationale for each function of Süzgeç. We compare the options using the seven criteria in the guidelines (cost, ease of use, scalability, performance, privacy, explainability, accessibility).

3.2 Overview

Süzgeç consists of six layers: (1) notification access (Android NotificationListenerService (Android Developers, n.d.-a): reads notifications, removes them from the panel, uses notification action buttons); (2) sound and silence (Do Not Disturb access: the phone stays silent, and Süzgeç plays the sound for Critical and Important); (3) classification (rule layer and Turkish text model); (4) personalisation (ranking that learns from user corrections); (5) on-device execution (LiteRT); (6) storage and scheduling (Room database, scheduled delivery). Development has three phases: phase 1 classifies notifications and removes them from the panel, phase 2 covers messages in SMS and e-mail apps, and phase 3 adds screen-time tracking, app transition analysis, a weekly report and time limits. The flow from a notification's arrival on the phone to the decision on its category is shown in Figure 19.

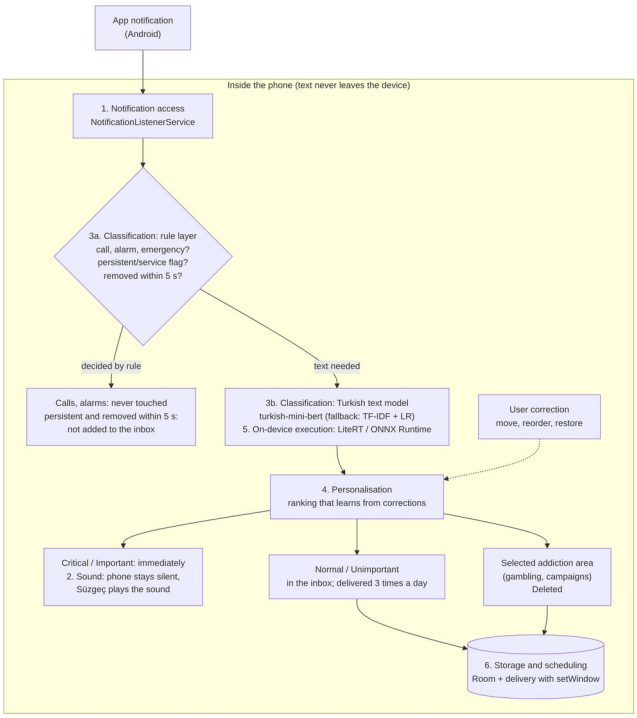


Figure 19. Süzgeç architecture (phase 1); the numbers in the boxes correspond to the six layers in 3.2: all processing happens on the phone; notifications the rule layer cannot decide go to the text model, and user corrections personalise the ranking (drawn by our team).

3.3 Relevance to the project

Table 12. Challenge, technology and basis

Challenge	Technology	Basis
Half of the notifications are persistent or short-lived	Android flags + 5-second wait (rule layer)	Finding 1 (2.5)
Importance depends on content; the text is Turkish	Turkish text classifier	Findings 4–6 (2.6); Perdana et al. (2024)
The profile varies between people	Ranking that learns from corrections	Finding 3; Lin, Y.-H. et al. (2024)
The model does not generalise to another text type	Retraining with team data and synthetic notifications	Finding 6 (Table 10)
Notification text contains private information	On-device model (LiteRT)	2.4 Ethics
Delivery 3 times a day and a silent phone	Scheduled delivery + Do Not Disturb access	Fitz et al. (2019); Visuri et al. (2019)

3.4 Comparison

Table 13. Text model options

Option	Cost and ease	Performance	Privacy and explainability	Decision
TF-IDF + logistic regression	Very low; trains quickly	Strong on short campaign texts (Perdana et al., 2024); in our own experiment macro F1 0.998 on SMS, about 0.5 ms per message (Table 10)	On device; word weights are readable	Baseline and fallback model
Small Turkish BERT	Medium; requires fine-tuning	Captures context better; expected to generalise better across text types, to be tested	On device; less explainable	Main model
Large language model (cloud)	High; online	Strong at personalisation (Zheng et al., 2025)	Text goes to a server	Will not be used

Other criteria: since the whole solution runs on the device, there is no server cost and cost does not grow with the number of users (scalability). Because the target is mid-range Android devices, the model will be kept small (accessibility); users will be able to see and change every decision (ease of use and explainability).

Tool and library choice: Besides the text model, we also compared options for the on-device runtime, the Turkish language model and delivery scheduling (Table 14).

Table 14. Tool and library options

Decision	Options	Comparison	Choice
On-device runtime	LiteRT (formerly TensorFlow Lite; Google) · ONNX Runtime Mobile (Microsoft) · ExecuTorch (Meta, PyTorch)	LiteRT is the most established option on Android and is backed by Google AI Edge documentation (Google AI Edge, n.d.); ONNX Runtime is framework-independent and can also run converted scikit-learn models (ONNX Runtime, n.d.); ExecuTorch is newer, for PyTorch models, with Android back-ends for CPU, GPU and Qualcomm, MediaTek and Samsung Exynos processors (Nachin et al., 2026)	LiteRT for the main model; ONNX Runtime for the TF-IDF + LR fallback model
Turkish language model	BERTurk · DistilBERTurk · small Turkish BERT family (turkish-mini-bert etc.)	BERTurk (about 110 million parameters) performs best but is large for a phone; DistilBERTurk is distilled from it and slightly behind (sentiment analysis macro F1 79.12 vs. 80.65; Schweter, 2020); turkish-mini-bert has only 11.6 million parameters and reaches 0.851 accuracy on sentiment analysis (BERTurk 0.919; Kesgin et al., 2023)	Main model turkish-mini-bert; compare with DistilBERTurk and BERTurk
Delivery scheduling	Exact alarm (AlarmManager) · time window (setWindow) · WorkManager	On Android 14 the exact alarm permission is denied by default for most newly installed apps; setWindow works with a window of at least 10 minutes; WorkManager periodic work runs at intervals of at least 15 minutes with no exact-time guarantee (Android Developers, n.d.-b)	setWindow with a window such as 09:00–09:10; exact alarm if the user grants permission
Storage	Room (SQLite) · DataStore	Room suits relational queries and history; DataStore suits small settings	Notifications in Room, settings in DataStore

3.5 Use cases

Table 15. Existing solutions and Süzgeç

Solution	What it does	Gap compared with Süzgeç
Android notification settings and Do Not Disturb	Turning off by app or channel, quiet hours	Manual and app-level; does not look at content, no batching
Pixel AI notification summaries	Summarises long message threads on the device; separating low-priority notifications is planned	Pixel 9 and 10 only; no Turkish and no addiction filter
NotificationDiary (Visuri et al., 2019)	On-device importance prediction, show/hide	Research prototype; no batching
NotiManager (Lin, Y.-H. et al., 2024)	Sorting, categories, pinning	Does not learn from corrections
one sec (Grüning et al., 2023)	Delay at app launch	Does not manage notifications
Yeşilay YEDAM 115	Free counselling for gambling and technology addiction	Not a technical tool; Süzgeç refers to it on its help page

3.6 Gaps and risks

Table 16. Risks and mitigations

Risk	Mitigation
Missing a Critical notification	Uncertain notifications go to the higher category; calls, alarms and emergency alerts are never touched
If Süzgeç stops, the phone stays silent	The previous sound setting is restored; pause button
The inbox turning into a new checking habit	Visits outside delivery times are counted and reported
Wrong deletion	Restorable from the Deleted box; learns from the correction
Privacy	All processing on the device; text never leaves the phone
Manufacturer battery restrictions may stop the service	Exemption from battery optimisation is requested; service status is monitored

3.7 Conclusion

Table 17. Layer and choice

Layer	Choice
Notification access	NotificationListenerService
Sound and silence	Do Not Disturb access; sound from Süzgeç
Classification	Rule layer + turkish-mini-bert (fallback: TF-IDF + logistic regression)
Personalisation	Ranking that learns from corrections
On-device execution	LiteRT (fallback model: ONNX Runtime)
Storage and scheduling	Room + DataStore; scheduled delivery with setWindow

3.8 References

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